FAST-CP: A Lightweight Perturbation-Aware Conformal Prediction Framework for Corrupted Time-Series Classification

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Abstract

Time-series classifiers deployed in sensor-driven environments often encounter corrupted inputs such as missing segments, local noise, and amplitude drift. These corruptions can make prediction sets from standard post-hoc uncertainty methods overly large or poorly matched to the observed input condition. This paper presents FAST-CP, a lightweight post-hoc conformal prediction framework for corrupted time-series classification under limited compute. The method computes cheap perturbation fingerprints, estimates a fingerprint-weighted local threshold, and mixes it with an augmented global conformal threshold. This design supports two practical operating modes: a coverage-prioritized mode that falls back to global augmented calibration, and a Pareto-oriented efficient mode that accepts a small measured coverage cost in exchange for smaller prediction sets. On 35 small-to-medium UCR datasets with five random seeds and three corruption families, the efficient setting reduces average prediction-set size by 3.02–4.44% relative to augmented split conformal prediction at $\alpha = 0.05$, with a mean coverage decrease of 0.17–0.86 percentage points. Additional UEA multivariate, MiniROCKET, and FCN checks show that the qualitative tradeoff is not tied to a single backbone. The full benchmark with control ablations completes in 47.7 minutes on an Apple M4 with 16 GB unified memory. FAST-CP is therefore intended as a reproducible, deployable coverage-efficiency tradeoff layer, not as a new conformal-validity guarantee under distribution shift.

Keywords: conformal prediction; time-series classification; uncertainty quantification; corrupted data; reproducible machine learning; pattern recognition

1 Introduction

Time-series classifiers are often evaluated on clean benchmark splits, but deployed sensors and monitoring systems rarely provide such inputs. Gaps, local noise, amplitude drift, and other benign corruptions can change the reliability of a classifier even when the target label remains meaningful. For applications that need uncertainty estimates rather than only top-1 predictions, this creates a practical problem: prediction sets should remain informative under corruptions, but should not grow so large that they lose decision value.

Conformal prediction provides a simple and model-agnostic way to convert classifier scores into prediction sets with finite-sample coverage guarantees under exchangeability [Vovk et al., 2005, Angelopoulos and Bates, 2023]. Recent work has studied conformal prediction for out-of-distribution time-series classification [Nanopoulos and Buza, 2025] and robust time-series classification with conformalized randomized smoothing [Franco et al., 2024]. These approaches highlight the same tension: robustness and coverage are valuable, but repeated smoothing inference or overly conservative global calibration may be unattractive when experiments and deployment must run on commodity hardware.

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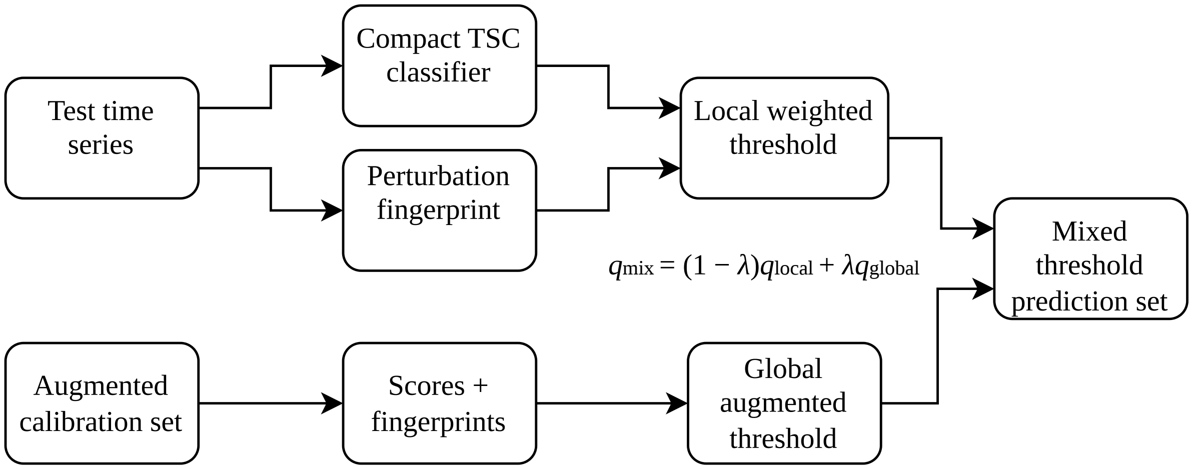

Figure 1: Overview of FAST-CP. A test time series is passed through a compact classifier and a perturbation-fingerprint extractor. Calibration examples with similar fingerprints define a local weighted conformal threshold, which is mixed with a global augmented threshold to form the final prediction set.

We study a smaller but practical question: can cheap descriptors of input perturbations expose a useful and controllable coverage-efficiency tradeoff for corrupted time-series conformal prediction? We answer this question with FAST-CP, a post-hoc conformal prediction framework for time-series classification. FAST-CP computes a perturbation fingerprint from each time series, including missingness, local variation, spectral entropy, autocorrelation, amplitude drift, and classifier confidence features. At test time, calibration examples with similar fingerprints receive larger weights when estimating a local conformal threshold. The local threshold is then mixed with an augmented global threshold, allowing the same framework to operate either as a coverage-prioritized global fallback or as a Pareto-oriented efficient mode.

We focus on three practical questions:

RQ1 Can perturbation-aware local weighting reduce conformal prediction-set size under common time-series corruptions?

RQ2 How large is the empirical coverage cost, and when should the method fall back to global augmented calibration?

RQ3 Is the method lightweight enough to be evaluated and deployed on commodity hardware?

Our contributions are:

1. We introduce a lightweight post-hoc conformal prediction framework for corrupted time-series classification, which adapts thresholds using interpretable perturbation fingerprints without retraining the base classifier.

2. We define a two-mode operating protocol that distinguishes coverage-prioritized deployment from Pareto-oriented efficient deployment. The former can fall back to augmented split conformal prediction, while the latter explicitly trades small empirical coverage loss for smaller prediction sets.

3. We provide a systematic corrupted time-series benchmark over 35 UCR datasets, five random seeds, three corruption families, and multiple baselines including APS, RAPS, confidence-weighted CP, and Mondrian CP.

4. We add external and backbone robustness checks on UEA multivariate datasets, MiniROCKET, and FCN classifiers, showing that the observed tradeoff is not limited to the compact random-convolution backbone.

5. We report practical runtime and reproducibility evidence showing that the full benchmark and ablations can be completed under modest compute.

The scope is deliberately practical. FAST-CP is not a new distribution-shift validity theorem; it is a reproducible Pareto calibration layer for corrupted time-series benchmarks, where the main question is whether a transparent post-hoc tradeoff is useful under modest compute.

2 Related Work

Time-series classification benchmarks. The UCR archive is a standard benchmark suite for univariate time-series classification, with a broad collection of small-to-medium datasets used across many methodological studies [Dau et al., 2019]. Python toolkits such as aeon provide practical infrastructure for time-series learning and archive access [Middlehurst et al., 2024]. Recent model work continues to explore compact convolutional, transformer, and hybrid architectures for time-series classification [Lan et al., 2024]. Our work does not aim to introduce a new classifier. Instead, it wraps a compact classifier with a post-hoc uncertainty layer and focuses on the calibration behavior under corruptions.

Conformal prediction. Conformal prediction constructs prediction sets from nonconformity scores and calibration data [Vovk et al., 2005]. Its appeal in modern machine learning is that it can wrap black-box predictors and provide distribution-free coverage under exchangeability [Angelopoulos and Bates, 2023]. In classification, prediction-set size is a natural efficiency measure: a method that attains the target coverage with smaller sets is more informative. Adaptive prediction sets and regularized adaptive prediction sets change the classification score to a cumulative probability score and are strong natural baselines for efficient conformal classification [Romano et al., 2020, Angelopoulos et al., 2021]. Mondrian conformal prediction conditions calibration on a discrete taxonomy, providing a useful contrast to our soft fingerprint weighting. Our method uses the standard split-conformal classification score $1 - p_\theta(y \mid x)$, but changes how the calibration threshold is estimated for corrupted test examples.

Weighted and local conformal methods. Our soft weighting is closely related to weighted conformal prediction under covariate shift and localized conformal prediction [Tibshirani et al., 2019, Guan, 2023]. Those lines of work clarify that weighting can be principled when the relevant density ratio or localization rule is well specified. FAST-CP is more modest: the weights are heuristic similarities in a perturbation-fingerprint space, not estimated likelihood ratios, and therefore they do not inherit the coverage guarantees of covariate-shift conformal prediction. This distinction is central to our framing.

Corrupted and out-of-distribution time series. Nanopoulos and Buza study conformal prediction for out-of-distribution time-series classification with distortions such as gaps [Nanopoulos and Buza, 2025]. Their work motivates the specific problem setting we consider: common production distortions can make neural classifiers overconfident or poorly calibrated. Our contribution is complementary. Rather than focusing only on global calibration variants, we compute perturbation fingerprints and use them to locally weight calibration examples.

Robust conformal methods. Franco et al. combine conformal prediction with randomized smoothing for robustness against real-world time-series perturbations [Franco et al., 2024]. This line of work is stronger when formal robustness arguments or smoothing certificates are required. FAST-CP targets a different regime: fast post-hoc calibration under limited compute. It does not replace certified smoothing methods, but provides a cheaper empirical alternative when repeated transformed inference is not practical.

Adaptive inference and uncertainty. Early-exit networks and nested prediction sets study uncertainty across intermediate exits of a model [Jazbec et al., 2024]. Such work modifies the architecture or analyzes sequential predictions within a network. FAST-CP is orthogonal: it leaves the classifier fixed and adapts only the conformal calibration threshold based on observed perturbation fingerprints.

3 Method

3.1 Problem setup

Let $x \in \mathbb{R}^T$ denote a univariate time series and $y \in \mathcal{Y}$ its class label. A trained classifier produces probabilities $p_\theta(y \mid x)$ over labels. Given calibration examples $\{(x_i, y_i)\}_{i=1}^n$, split conformal classification commonly uses the nonconformity score

$$s(x_i, y_i) = 1 - p_\theta(y_i \mid x_i). \tag{1}$$

For a target miscoverage level α, a global conformal threshold q_{global} is computed as the empirical $(1-\alpha)$ quantile with the usual finite-sample correction. The prediction set for a test input x is

$$\mathcal{C}(x) = \{y \in \mathcal{Y} : 1 - p_\theta(y \mid x) \leq q_{\text{global}}\}. \tag{2}$$

Under exchangeability between calibration and test examples, this global split-conformal procedure has the standard finite-sample marginal coverage guarantee. Our method keeps the classifier fixed and modifies only the calibration threshold. Under corrupted evaluation inputs, the aim is to expose a practical coverage-efficiency tradeoff rather than to restore distribution-free validity under arbitrary shift.

For evaluation, we report empirical coverage, average prediction-set size, and absolute coverage gap. For a collection of test examples $\mathcal{D}$, the absolute coverage gap is

$$\text{AbsGap}(\mathcal{D}) = \left| (1-\alpha) - \frac{1}{|\mathcal{D}|} \sum_{(x,y)\in\mathcal{D}} \mathbf{1}\{y \in \mathcal{C}(x)\} \right|. \tag{3}$$

When comparing two methods, an absolute-gap delta is the candidate method's absolute gap minus the baseline method's absolute gap; negative values indicate that the candidate is closer to the nominal target.

3.2 Perturbation fingerprints

FAST-CP represents each time series by a fingerprint $\phi(x)$ designed to be cheap to compute and interpretable. The current implementation uses missingness, local variation, spectral entropy, lag-one autocorrelation, amplitude drift, peak-to-standard-deviation ratio, top-class confidence, and logit margin. These features are not meant to be a learned representation of the label; they summarize how the input looks from a calibration-risk perspective.

We also use augmented calibration data. For each calibration series, we generate corrupted variants using the same corruption families as evaluation: missing segments, additive sensor noise, and amplitude drift. The classifier is not retrained on these variants; they are used only to populate the calibration score and fingerprint pool.

3.3 Local weighted conformal threshold

For a test input x, calibration example i receives a similarity weight

$$w_i(x) = \exp\left(-\frac{\|\tilde{\phi}(x_i) - \tilde{\phi}(x)\|_2^2}{2h^2}\right), \tag{4}$$

where $\tilde{\phi}$ denotes standardized fingerprints and h is set by a median-distance heuristic. The local threshold $q_{\text{local}}(x)$ is the weighted empirical $(1-\alpha)$ quantile of calibration nonconformity scores.

In the implementation, scores are sorted increasingly and their nonnegative weights are normalized to sum to one. The weighted quantile is the first sorted score whose cumulative normalized weight is at least $1-\alpha$. Ties are therefore handled by the sorted-score order and the threshold is conservative with respect to the weighted empirical CDF. Unlike the unweighted split-conformal threshold, this weighted threshold does not use the exact finite-sample exchangeability correction because the soft weights are heuristic and test-dependent.

Pure local weighting can produce thresholds that are too aggressive when the local neighborhood is small or noisy. We therefore use a local-global threshold family,

$$q_{\text{mix}}(x) = (1-\lambda)q_{\text{local}}(x) + \lambda q_{\text{global}}, \tag{5}$$

where $\lambda \in [0, 1]$ controls the local-global tradeoff. $\lambda = 0$ recovers purely local FAST-CP, while $\lambda = 1$ recovers augmented split conformal prediction because the local threshold is ignored.

The final prediction set is

$$\mathcal{C}_{\text{FAST}}(x) = \{y \in \mathcal{Y} : 1 - p_\theta(y \mid x) \leq q_{\text{mix}}(x)\}. \tag{6}$$

3.4 Operating modes

FAST-CP is not intended to replace the global conformal threshold in all settings. Instead, it defines a local-global threshold family with two deployment modes.

FAST-CP-Safe / coverage-prioritized mode. This mode selects λ by validation under an empirical coverage constraint. If no intermediate λ satisfies the target tolerance, the method falls back to $\lambda = 1$, which is exactly augmented split conformal prediction on the augmented calibration pool. This mode is intended for settings where coverage is the primary objective and smaller prediction sets are secondary.

FAST-CP-Efficient / Pareto-oriented mode. This mode uses an intermediate λ as a Pareto operating point when a small empirical coverage decrease is acceptable in exchange for smaller prediction sets. In the main benchmark we report $\lambda = 0.35$ as a fixed Pareto point rather than as an independently validated default. The independent validation experiment in Figure 5 and Table 7 is therefore part of the method's deployment logic: it tells the user whether to operate at an intermediate point or revert to the global fallback.

In practical use, we recommend the following deployment protocol. First, construct an augmented calibration pool using the corruption types expected in the deployment environment. Second, compute perturbation fingerprints and evaluate a small grid of λ values on a validation split. Third, if empirical coverage is a hard requirement, select the smallest-set λ satisfying the coverage tolerance; otherwise use an intermediate λ as a Pareto-efficient operating point. Finally, monitor undercoverage and fall back to the global augmented threshold when corruption severity increases or validation coverage becomes unstable.

Algorithm 1 FAST-CP deployment protocol

Require: Trained classifier p_θ, calibration set, expected corruption families, target α, candidate grid Λ

1: Construct an augmented calibration pool using expected deployment corruptions.
2: Compute nonconformity scores and perturbation fingerprints for all calibration variants.
3: Compute the augmented global threshold q_{global}.
4: **for** each validation or test input x **do**
5: Compute $\phi(x)$ and the local weighted threshold $q_{\text{local}}(x)$.
6: Form $q_{\text{mix}}(x) = (1-\lambda)q_{\text{local}}(x) + \lambda q_{\text{global}}$.
7: Output $\mathcal{C}_{\text{FAST}}(x) = \{y : 1 - p_\theta(y \mid x) \le q_{\text{mix}}(x)\}$.
8: **end for**

9: **Coverage-prioritized use:** choose the smallest-set $\lambda \in \Lambda$ satisfying the validation coverage tolerance; otherwise set $\lambda = 1$.
10: **Pareto-oriented use:** choose an intermediate λ only when the measured coverage cost is acceptable.

Table 1: Recommended operating regimes for the local-global threshold family.

Scenario	Recommended mode	Reason
Coverage-critical deployment	FAST-CP-Safe / Aug SCP fallback	Preserve global augmented calibration
Low-to-moderate corruption	FAST-CP-Efficient	Smaller sets with measured coverage cost
Severe corruption	Global fallback	Both methods are already strained
Limited compute	Post-hoc FAST-CP layer	No retraining or repeated smoothing inference
Unknown deployment shift	Validate before selecting λ	No arbitrary-shift coverage guarantee

4 Experiments

4.1 Setup

We evaluate on 35 small-to-medium UCR time-series classification datasets selected by a fixed rule before inspecting method outcomes. Starting from a broad UCR candidate list, we keep datasets with at most 2500 total examples, sequence length at most 512, and at most 20 classes. One initially selected dataset, DiatomSizeReduction, was excluded because some train/calibration splits left a class absent from the model-fitting split; it was replaced by the next eligible dataset, MoteStrain. The full selection table, including excluded datasets and reasons, is saved in the experiment artifacts.

We consider three corruption families: missing segments (gap), additive Gaussian sensor noise (noise), and multiplicative amplitude drift (drift). Each corruption is evaluated at three severity levels. Unless otherwise stated, we report results at $\alpha = 0.05$ with seeds 0–4.

We evaluate FAST-CP as a post-hoc decision layer rather than as a replacement for exchangeability-based conformal validity. The base classifier uses random convolutional features followed by logistic regression. To keep the expanded run tractable on Apple M4, the 35-dataset experiment uses 64 random kernels. We compare clean split conformal prediction, augmented split conformal prediction, label-smoothed augmented conformal prediction, confidence-weighted conformal prediction, APS, RAPS, a corruption-family Mondrian conformal baseline, pure local FAST-CP, and the mixed-threshold FAST-CP variants. Clean split conformal prediction calibrates only on the original calibration split. Augmented split conformal prediction pools corrupted calibration variants and applies one global threshold; it is also the global fallback selected

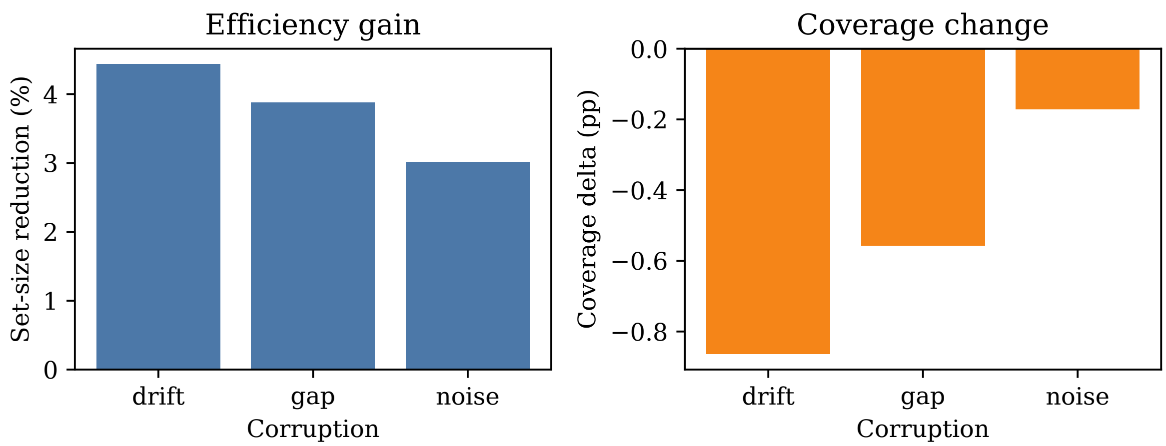

Figure 2: Expanded 35-dataset, five-seed benchmark at $\alpha = 0.05$. FAST-CP-efficient reduces prediction-set size relative to augmented split conformal prediction across drift, gap, and noise corruptions, with a small measured coverage decrease.

when coverage-prioritized validation chooses $\lambda = 1$. FAST-CP and confidence-weighted CP use this same augmented calibration pool for local weighting; they do not weight only the clean calibration examples. Label-smoothed augmented conformal prediction applies a small probability smoothing transform before computing scores. Confidence-weighted conformal prediction uses the same local weighting machinery as FAST-CP, but only with top-class confidence and logit-margin features rather than the full perturbation fingerprint. The Mondrian baseline conditions its threshold on the known corruption family, making it a strong grouped-calibration comparison.

The efficient operating point fixes the local-global mixing coefficient to $\lambda = 0.35$ after the five-dataset ablation in Figure 4. The expanded 35-dataset, five-seed comparison is therefore reported with one fixed setting rather than tuning λ separately for each dataset or corruption. The independent validation protocol below determines whether this point should be treated as a default or only as a Pareto choice.

To check whether the effect is tied to the compact random-convolution/logistic backbone, we also run two supplementary backbone checks. The first uses MiniROCKET features [Dempster et al., 2021] with a calibrated ridge classifier on 15 UCR datasets. The second uses an FCN time-series classifier on eight representative UCR datasets with two seeds and 20 training epochs. These checks use the same conformal and fingerprint pipeline, so only the base time-series classifier changes.

4.2 Main coverage-efficiency tradeoff

Figure 2 summarizes the main comparison from the expanded benchmark. FAST-CP-efficient reduces average prediction-set size by 4.44% under drift, 3.87% under gaps, and 3.02% under noise relative to augmented split conformal prediction. The corresponding coverage deltas are -0.0086, -0.0056, and -0.0017, respectively. Thus, the larger benchmark confirms a statistically significant efficiency gain, but also shows that the gain is not free: coverage decreases slightly on average, especially under drift.

The absolute coverage-gap delta is near zero under drift, slightly negative under gaps, and negative under noise. The overall mean absolute-gap delta is -0.0012 with a 95% bootstrap confidence interval of [-0.0022, -0.0002]. We therefore frame the method as a significant set-size reduction with a small measured coverage tradeoff, not as a uniformly better calibrated method.

Table 2: Coverage-efficiency comparison under corrupted UCR evaluation. Size Δ is relative to augmented split conformal prediction. Positive values indicate smaller sets than Aug SCP, but should be interpreted jointly with coverage. FAST-CP-efficient is a Pareto point, not a coverage-dominant method.

Method	Coverage	Set size	Abs. gap	Size Δ vs Aug	Drift size	Gap size	Noise size
Clean SCP	0.8656	1.9323	0.1049	0.4862	1.9412	1.9240	1.9315
Aug SCP / global fallback	0.9501	2.4185	0.0467	0.0000	2.4405	2.4151	2.3998
APS	0.9757	3.2595	0.0423	-0.8411	3.2654	3.2128	3.3005
RAPS	0.9730	2.7313	0.0424	-0.3128	2.7262	2.7293	2.7384
Mondrian	0.9553	2.5021	0.0418	-0.0837	2.3172	2.7412	2.4480
Conf-weighted	0.9478	2.4262	0.0440	-0.0078	2.4421	2.4310	2.4056
FAST-CP-local	0.9440	2.3112	0.0446	0.1073	2.3003	2.3096	2.3236
FAST-CP-efficient	0.9448	2.3254	0.0455	0.0931	2.3309	2.3240	2.3213

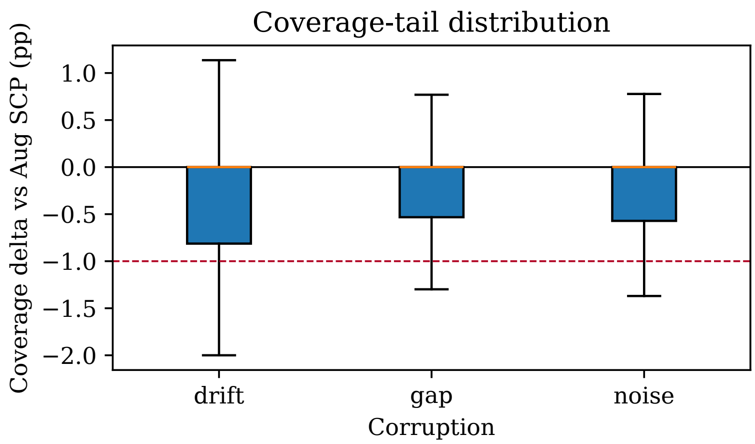

Figure 3: Distribution of coverage deltas for FAST-CP-efficient relative to augmented split conformal prediction. The dashed line marks a one percentage point decrease.

4.3 Tail-risk analysis and deployment boundary

Figure 3 and Table 3 show the tail behavior behind the average tradeoff. FAST-CP-efficient increases the overall fraction of evaluation cells below nominal coverage from 32.83% to 39.05%, and below nominal by more than one percentage point from 26.98% to 31.11%. The worst ten coverage drops are concentrated in very small test sets such as Coffee and BirdChicken, where one or two examples move empirical coverage substantially. Confidence-only weighting is safer in this audit, with mean coverage 0.9494 and a smaller undercoverage tail, but its prediction sets are much closer in size to augmented split conformal prediction.

FAST-CP-efficient is therefore not recommended when coverage is a hard safety constraint. In such settings, the coverage-prioritized mode should be used; on this benchmark that mode falls back to the augmented global threshold. This is the intended deployment boundary of the local-global family rather than an exception to the method.

4.4 Severity-level analysis

Table 4 shows that severe corruptions are difficult for both methods: augmented split conformal prediction already falls to 0.9058 mean coverage at the high severity level. The FAST-CP coverage drop remains on the same order across severities, while the set-size reduction becomes smaller under high severity. The practical

Table 3: Overall undercoverage tail-risk audit. Rates are fractions of the 1575 dataset/seed/corruption/severity evaluation cells.

Method	Mean cov.	Below target	Below target -1pp	P10 cov.
Aug SCP / global fallback	0.9501	0.3283	0.2698	0.8777
FAST-CP-efficient	0.9448	0.3905	0.3111	0.8699
Confidence-only	0.9494	0.3283	0.2660	0.8847

Table 4: Severity-level behavior averaged over gap, noise, and drift corruptions. Low, medium, and high correspond to the three severity levels within each corruption family.

Severity	Aug cov.	FAST cov.	Cov. delta	Set reduction
Low	0.9774	0.9730	-0.0044	0.1115
Medium	0.9671	0.9607	-0.0064	0.1017
High	0.9058	0.9007	-0.0052	0.0661

use case of FAST-CP-efficient is therefore low-to-moderate corruption, where the global threshold is not already severely strained. Under severe corruptions, users should prefer validation-selected global fallback behavior.

4.5 Mismatched corruption stress test

The main benchmark augments calibration with the same corruption families used at evaluation time. To test a less favorable deployment setting, we run a mismatched stress test on 10 representative UCR datasets with three seeds. The augmented calibration pool contains only gap and noise variants, while test inputs are evaluated under drift and mixed corruptions. The mixed corruption applies a missing segment, additive noise, and amplitude drift sequentially. This experiment is intentionally a stress test rather than a new validity claim: it asks whether the local-global threshold family remains useful when the calibration corruption menu is incomplete.

Table 5 shows that the efficiency trend persists under mismatch, but the operating boundary becomes clearer. On drift, where the global augmented threshold remains near nominal coverage, FAST-CP-efficient reduces set size by 0.1252 labels while decreasing coverage by 0.57 percentage points. On mixed corruptions, both methods under-cover substantially, with augmented split conformal prediction at 0.8969 coverage and FAST-CP-efficient at 0.8929. The correct interpretation is therefore not that FAST-CP solves unseen corruptions; rather, mismatch should trigger validation and possible global fallback, especially when the global threshold itself is already below target.

4.6 External multivariate validation

To reduce dependence on univariate UCR evidence, we add a small external validation on three UEA multivariate datasets: BasicMotions, Epilepsy, and NATOPS. These datasets contain real multichannel sensor or motion signals; we evaluate them under synthetic gap, noise, and drift stress tests. The run uses three seeds, $\alpha = 0.05$, 24 random convolution kernels per channel, and the same augmented calibration pool and $\lambda = 0.35$ mixed threshold.

Table 6 gives a small but useful external check. The set-size reduction remains positive across all three corruption families, while the overall coverage delta is -0.0009 and its bootstrap confidence interval includes zero. This does not replace real corrupted deployment data, but it shows that the basic tradeoff is not limited to univariate UCR series.

Table 5: Mismatched corruption stress test. Calibration augmentation uses gap and noise only; test inputs use drift or mixed corruptions. Values compare FAST-CP-efficient with augmented split conformal prediction across 180 dataset/seed/corruption/severity cells.

Test corruption	Aug cov.	FAST cov.	Cov. delta	Set reduction
Drift	0.9549	0.9492	-0.0057	0.1252
Mixed	0.8969	0.8929	-0.0040	0.0620
Overall	0.9259	0.9211	-0.0048	0.0936

Table 6: External UEA multivariate validation. Values compare the mixed FAST-CP setting with augmented split conformal prediction across 81 dataset/seed/corruption/severity cells.

Corruption	FAST cov.	Aug cov.	Cov. delta	Set reduction
Drift	0.9499	0.9519	-0.0019	0.0441
Gap	0.9491	0.9479	0.0012	0.0527
Noise	0.9639	0.9658	-0.0019	0.0542
Overall	0.9543	0.9552	-0.0009	0.0503

4.7 Local-global mix ablation

Figure 4 shows that the mixing coefficient is not merely cosmetic. Pure local weighting gives smaller sets but weaker coverage behavior, while $\lambda = 1$ recovers augmented split conformal prediction with larger sets. Intermediate values, particularly 0.2 and 0.35, provide practical tradeoff points on the ablation subset.

4.8 Validation of safe fallback and Pareto operating points

To validate the two-mode protocol, we split the 35 valid UCR datasets into 10 tuning datasets and 25 held-out evaluation datasets. On the tuning split, we evaluate $\lambda \in \{0, 0.1, 0.2, 0.35, 0.5, 0.65, 0.8, 1\}$ at $\alpha \in \{0.05, 0.1, 0.2\}$ with three seeds. A lambda is considered feasible only if its mean coverage is at least $1 - \alpha - 0.01$ for every alpha; among feasible lambdas, the rule chooses the smallest average set size.

No lambda satisfied this feasibility rule on the tuning split. The fallback rule, which minimizes total coverage shortfall and then set size, selected $\lambda = 1$, equivalent to augmented split conformal prediction. This result confirms the need to distinguish the two operating modes. When coverage is prioritized, the validation protocol selects the global augmented threshold. When efficiency is prioritized, intermediate λ values define Pareto points with measurable set-size reductions and explicit coverage costs.

Figure 5 summarizes the held-out Pareto behavior. At $\alpha = 0.05$, $\lambda = 0.35$ reduces average set size by 0.1125 labels relative to $\lambda = 1$ with a 95% bootstrap confidence interval of [0.0997, 0.1269], but lowers coverage by 0.0083. At $\alpha = 0.10$, the reduction decreases to 0.0392 labels with a coverage delta of -0.0052. At $\alpha = 0.20$, the reduction is only 0.0109 labels and the coverage delta is not statistically clear. Thus, local weighting is most useful at stricter coverage levels, while broader alpha settings make the efficiency advantage much smaller.

4.9 MiniROCKET backbone check

As a stronger-backbone supplement, we repeat the evaluation on 15 fixed UCR datasets with MiniROCKET features and a calibrated ridge classifier. The run uses three seeds, $\alpha \in \{0.05, 0.10\}$, and the same gap, noise, and drift corruptions. Compared with $\lambda = 1$, the $\lambda = 0.35$ mixed threshold reduces average set size by 0.0943 labels at $\alpha = 0.05$ with a 95% confidence interval of [0.0770, 0.1121], while decreasing coverage

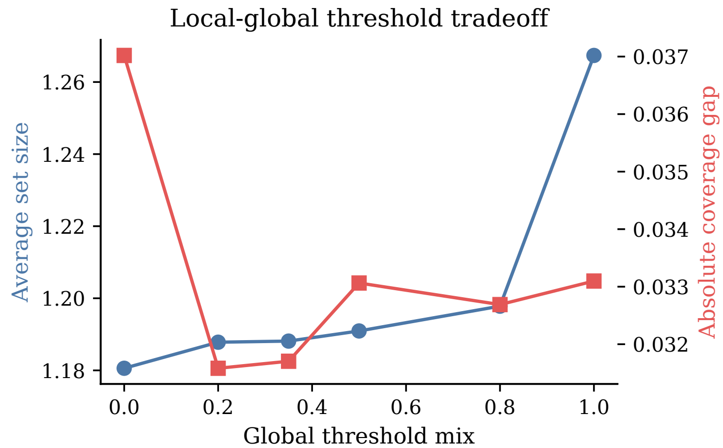

Figure 4: Ablation of the local-global threshold mix on five datasets. Intermediate mixing recovers coverage relative to purely local thresholds while retaining smaller prediction sets than the global augmented threshold.

Table 7: Held-out tradeoff of $\lambda = 0.35$ relative to $\lambda = 1$ (Aug SCP) across target miscoverage levels.

α	Set reduction	95% CI	Coverage delta	95% CI
0.05	0.1125	[0.0997, 0.1269]	-0.0083	[-0.0109, -0.0058]
0.10	0.0392	[0.0296, 0.0499]	-0.0052	[-0.0083, -0.0026]
0.20	0.0109	[0.0046, 0.0179]	-0.0009	[-0.0032, 0.0012]

by 0.0055. At $\alpha = 0.10$, it reduces set size by 0.0694 labels with a 95% confidence interval of [0.0559, 0.0833], while decreasing coverage by 0.0067. The qualitative tradeoff therefore persists under a stronger MiniROCKET-based classifier, although this does not establish transfer to deep InceptionTime, ResNet, or transformer backbones.

4.10 FCN backbone check

We also run a small deep-backbone check with an FCN classifier on eight representative UCR datasets: ECG200, GunPoint, Coffee, BME, FaceFour, ItalyPowerDemand, MoteStrain, and Beef. The run uses two seeds, 20 epochs, $\alpha = 0.05$, and the same gap, noise, and drift corruptions. This is not a full deep TSC benchmark, but it directly tests whether the mixed-threshold behavior disappears once the base classifier is no longer a linear model over random features.

Table 8 shows a smaller but consistent efficiency gain under FCN. The overall set-size reduction is 0.0139 labels with a 95% bootstrap confidence interval of [0.0090, 0.0194], while the coverage delta is -0.0014 with a 95% interval of [-0.0024, -0.0006]. This supports the narrower claim that the Pareto behavior is not exclusive to the compact baseline, while leaving larger InceptionTime, ResNet-style, and transformer TSC evaluations as future work.

4.11 Dataset-level behavior

At the dataset/seed/corruption/severity level, FAST-CP-efficient obtains a set-size win rate of 71.30%. Its coverage is within one percentage point of augmented split conformal prediction in 73.33% of comparison cells, and its absolute coverage-gap win rate is 82.48%. These results show that the aggregate gains are not

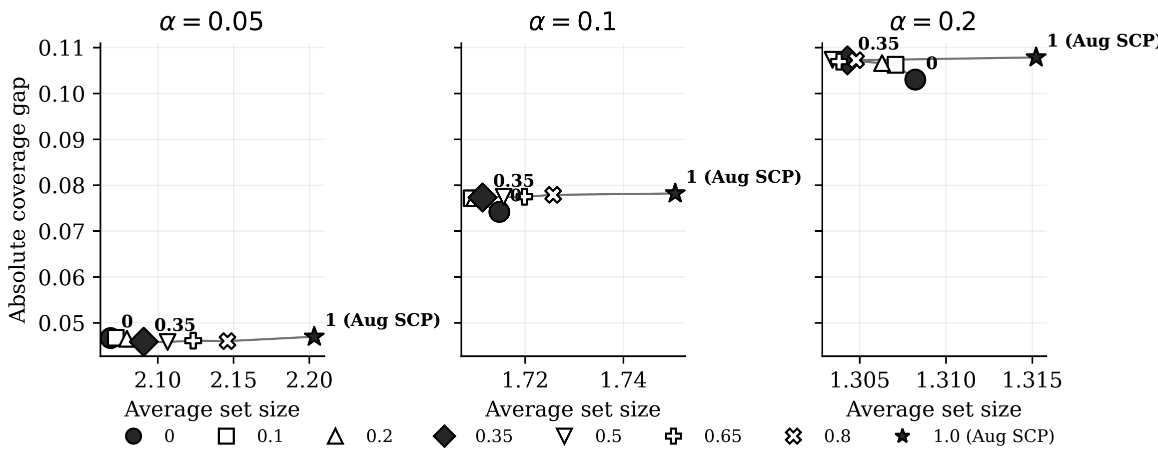

Figure 5: Held-out alpha/lambda Pareto analysis over 25 evaluation datasets. Each panel averages over corruption families and plots the local-global threshold path from pure local weighting ($\lambda = 0$) to augmented split conformal prediction ($\lambda = 1$, labeled as Aug SCP).

Table 8: Small FCN backbone check. Values compare the mixed FAST-CP setting with augmented split conformal prediction across 144 dataset/seed/corruption/severity cells.

Corruption	FAST cov.	Aug cov.	Cov. delta	Set reduction
Drift	0.9519	0.9544	-0.0026	0.0139
Gap	0.9426	0.9431	-0.0005	0.0139
Noise	0.9679	0.9691	-0.0012	0.0138
Overall	0.9541	0.9556	-0.0014	0.0139

driven solely by one corruption type, although several datasets remain unfavorable for set-size reduction.

4.12 Statistical tests and stronger baselines

To avoid over-emphasizing the 1575 evaluation cells as if they were independent datasets, we also report a dataset-level block bootstrap. Each dataset is first averaged over seeds, corruptions, and severities, and the bootstrap resamples datasets. Under this more conservative summary, FAST-CP-efficient reduces set size by 0.0931 labels with a 95% dataset-block confidence interval of [0.0646, 0.1236]; 32 of 35 datasets have a positive average set-size reduction. The corresponding dataset-block coverage delta is -0.0053 with interval [-0.0086, -0.0025], while the absolute-gap delta interval [-0.0029, 0.0006] overlaps zero. Cell-level paired tests lead to the same qualitative conclusion, but we treat them as supporting evidence rather than the main statistical claim.

Against APS and RAPS, FAST-CP-efficient produces much smaller sets, but with larger coverage drops; these baselines are more conservative on this benchmark. Against the corruption-family Mondrian baseline, FAST-CP-efficient has smaller sets overall but worse coverage and absolute-gap behavior under gap and noise. This comparison makes the tradeoff explicit: the method is efficient, but not uniformly dominant.

4.13 Fingerprint ablations

Table 9 clarifies the role of the fingerprint. The gain of FAST-CP comes from two sources: local-threshold variability induced by the mixed conformal family, and meaningful perturbation-aware weighting. The

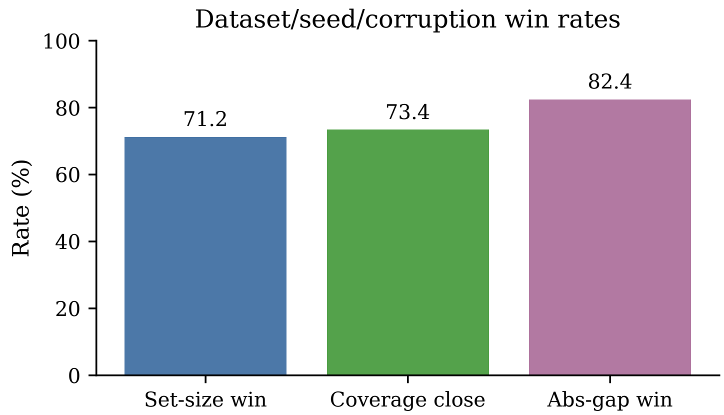

Figure 6: Dataset/seed/corruption-level comparison of FAST-CP-efficient against augmented split conformal prediction.

Table 9: Fingerprint contribution audit under the same mixed-threshold framework. All variants use the augmented calibration pool and $\lambda = 0.35$; size Δ is measured against augmented split conformal prediction.

Variant	Coverage	Set size	Abs. gap	Size Δ
Random fingerprint	0.9440	2.3681	0.0489	0.0504
Input-only	0.9387	2.2820	0.0474	0.1365
Confidence-only	0.9494	2.3921	0.0443	0.0263
Full fingerprint	0.9448	2.3254	0.0455	0.0931

random-fingerprint control shows that the threshold family itself can reduce set size, but its 0.0504-label reduction is substantially smaller than the 0.0931-label reduction from the full fingerprint and comes with worse absolute coverage gap. Confidence-only weighting is safer, with mean coverage 0.9494 and the smallest undercoverage tail, but it gives only a 0.0263-label set-size reduction, so it behaves more like a conservative confidence-adaptive baseline than an efficiency-oriented setting. Input-only descriptors are aggressive, reducing set size by 0.1365 labels but lowering mean coverage to 0.9387. The full fingerprint is therefore not included because it dominates every alternative on every metric; it is included because it uses input perturbation descriptors to recover much of the input-only efficiency while using confidence and margin to temper the largest coverage loss. This balanced operating regime is the practical reason to prefer the full fingerprint when smaller sets are desired but the input-only variant is too aggressive and the confidence-only variant is too close to the global fallback.

5 Runtime and Practical Analysis

Figure 7 and Table 10 report wall-clock runtime on Apple M4 16 GB. The 35-dataset, five-seed benchmark with fingerprint-control ablations completes in 2863.67 seconds, or 47.7 minutes, including APS/RAPS, Mondrian, random-fingerprint, input-only, and confidence-only baselines. The average runtime is 16.4 seconds per dataset-seed pair. The independent alpha/lambda tuning sweep completes in 838.52 seconds, and the 25-dataset held-out alpha/lambda evaluation completes in 1190.63 seconds. These timings include dataset loading, feature extraction, classifier fitting, corruption generation, and conformal evaluation.

The runtime profile supports the intended compute regime: the method can be evaluated end-to-end on commodity hardware rather than a dedicated GPU server. The main bottlenecks are random convolution

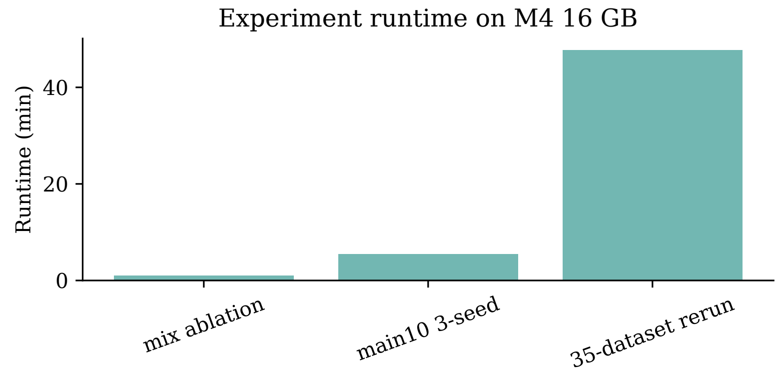

Figure 7: Runtime of the main and ablation experiments on Apple M4 16 GB.

Table 10: Runtime summary on Apple M4 16 GB. Times include dataset loading, feature extraction, model fitting, corruption generation, and conformal evaluation.

Run	Datasets	Seeds	Corrupt.	α	Rows	Min.	Sec./ds-seed
smoke	2	1	2	2	120	0.19	5.8
pilot5	5	1	3	2	450	1.42	17.0
pilot5_mixed	5	1	3	2	540	1.34	16.1
main10	10	1	3	1	540	1.94	11.7
mix_ablation5	5	1	3	1	495	1.01	12.1
main10_3seed	10	3	3	1	1620	5.43	10.9
ucr35_5seed_controls	35	5	3	1	25200	47.73	16.4
lambda_tune10	10	3	3	3	12960	13.98	28.0
lambda_eval25	25	3	3	3	32400	19.84	15.9
minirocket15	15	3	3	2	8100	0.37	0.5
uea_external	3	3	3	1	486	2.24	14.9
fcn8	8	2	3	1	576	0.76	2.9

feature extraction and repeated local weighted-threshold computation. This is acceptable for the small-to-medium UCR subset used here, but scaling to much larger datasets requires caching, vectorized feature extraction, or approximate nearest-neighbor weighting.

The practical lesson is that FAST-CP is best viewed as a post-hoc calibration layer. Once classifier probabilities and fingerprints are cached, the conformal baselines and local-global threshold variants are fast to recompute. This makes ablations over λ and corruption families cheap relative to training new neural architectures.

Reproducibility artifacts. All experiment scripts save raw per-cell results and aggregated summaries as CSV files. The project manifest and experiment log record commands, runtimes, output paths, and date-stamped updates, so the reported tables can be regenerated from saved artifacts.

Smoothing-proxy comparison. To quantify the cost of repeated transformed inference, we also ran a small smoothing-proxy baseline on ECG200, GunPoint, and Coffee with eight noisy evaluations per calibration or test example. This proxy obtains similar mean coverage to the mixed FAST-CP setting (0.9732 versus 0.9717), but its average prediction-set size is larger (1.4495 versus 1.2853). It also adds 11.8 seconds of repeated-inference overhead per corruption/severity evaluation cell in this small benchmark. While this is not a full reproduction of conformalized randomized smoothing, it supports the practical motivation for a post-hoc fingerprint method when compute is limited.

6 Limitations and Conclusion

The current evidence is deliberately scoped in three ways. First, the validity and calibration scope is empirical: FAST-CP is post-hoc, and local weighting with augmented calibration pooling does not restore finite-sample conformal guarantees under arbitrary distribution shift. The expanded benchmark therefore supports a coverage-efficiency tradeoff, not a free calibration improvement: prediction sets are significantly smaller, but mean coverage and undercoverage-tail behavior are worse than augmented split conformal prediction.

Second, the data and model coverage remains bounded. The strongest results are at $\alpha = 0.05$ on 35 small-to-medium univariate UCR datasets with synthetic corruptions. We add $\alpha \in \{0.1, 0.2\}$, held-out λ Pareto analysis, UEA multivariate validation, MiniROCKET, and FCN checks, but these supplements do not replace real corrupted deployment datasets or a full deep TSC study over InceptionTime, ResNet-style models, and transformers.

Third, the fingerprint and hyperparameter design is heuristic. Confidence and margin features are useful but may introduce circularity, because the classifier's own uncertainty helps determine the threshold. The bandwidth h and mix λ have no adaptive error-control rule; under independent coverage-prioritized validation, the selected value is $\lambda = 1$, so intermediate lambdas should be reported as Pareto tradeoff points rather than validated defaults. The random-fingerprint and feature-ablation controls further show that the full hand-designed fingerprint is not uniquely responsible for the efficiency gain: confidence-only weighting is safer, input-only weighting is more aggressive, and random weighting still contributes some set-size reduction.

Despite these limits, the results support a practical conclusion. Perturbation-aware local weighting, combined with global fallback calibration, provides a lightweight way to adapt conformal thresholds to corrupted time-series inputs. Coverage-prioritized validation preserves the augmented global threshold when coverage is the primary objective; the Pareto-oriented setting improves prediction-set efficiency with a measurable coverage cost. For practitioners working under modest compute budgets, FAST-CP offers a reproducible calibration layer when this tradeoff is acceptable.

Statements and Declarations

Code availability. The source code, reproduction scripts, public data access instructions, and saved result files are available at `https://github.com/zoumaomao/fast-cp-reproducibility`.

Data availability. This study uses public benchmark datasets from the UCR Time Series Classification Archive and the UEA Multivariate Time Series Classification Archive. The repository provides dataset access instructions through the `aeon` loaders, together with saved CSV/JSON/Markdown outputs supporting the reported results.

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Author contributions. Maomao Zou conceived the study, implemented the method, conducted the experiments, analyzed the results, prepared the reproducibility materials, and wrote and revised the manuscript.

Use of generative AI tools. During the preparation of this manuscript, the author used ChatGPT to assist with manuscript drafting, language editing, and improving the readability and organization of the text. The experimental design, implementation, data analysis, interpretation of results, and scientific conclusions were conducted and verified by the author. The author reviewed and edited all AI-assisted text and takes full responsibility for the accuracy, originality, and integrity of the final manuscript.

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A Implementation Details

The experimental code uses UCR text files downloaded from the time-series classification archive. Each series is z-normalized after simple interpolation of missing values. Random convolutional features are generated with kernel lengths 7, 9, and 11, followed by logistic regression. Corruptions are generated only for calibration augmentation and test evaluation; the classifier is not retrained on corrupted samples.

B Additional Notes

The smoothing-proxy experiment is included only as a small practical-cost comparison. It repeats noisy inference eight times per calibration or test example on ECG200, GunPoint, and Coffee, but it is not a full reproduction of conformalized randomized smoothing. The corresponding results are stored in `results/smoothing_proxy3_alpha005_method_summary.csv`.

C A Stability View of Local Thresholds

This appendix gives a small conditional stability statement for the local threshold used by FAST-CP. It is not a distribution-free validity theorem under shift, and it should not be read as evidence that the fingerprint weights are correct. Its purpose is narrower: if the weighted empirical score CDF already approximates an ideal local score CDF, then the resulting local quantile is stable. The assumption is strong and essentially encodes the desired localization quality.

For a test fingerprint z, define the weighted empirical score CDF

$$F_z(t) = \frac{\sum_{i=1}^{n} w_i(z)\mathbf{1}\{s_i \le t\}}{\sum_{i=1}^{n} w_i(z)}. \tag{7}$$

Let $q_z(\alpha) = \inf\{t : F_z(t) \ge 1-\alpha\}$ be the corresponding local weighted quantile. Let $F_\star$ denote an ideal local score CDF for examples from the same perturbation regime, with quantile $q_\star(\alpha)$.

Proposition 1 (Local quantile stability)**.** *Suppose* $\sup_t |F_z(t) - F_\star(t)| \le \epsilon$. *Assume that* $F_\star$ *has local mass at* $q_\star(\alpha)$*: there are constants* $m > 0$ *and* $r > 0$ *such that, for every* u *with* $|u - q_\star(\alpha)| \le r$*,*

$$F_\star(u) - F_\star(q_\star(\alpha)) \ge m(u - q_\star(\alpha)), \qquad u \ge q_\star(\alpha), \tag{8}$$

$$F_\star(q_\star(\alpha)) - F_\star(u) \ge m(q_\star(\alpha) - u), \qquad u \le q_\star(\alpha). \tag{9}$$

If $\epsilon/m \le r$*, then*

$$|q_z(\alpha) - q_\star(\alpha)| \le \epsilon/m. \tag{10}$$

For the mixed threshold $q_{\mathrm{mix}}(z) = (1-\lambda)q_z(\alpha) + \lambda q_{\mathrm{global}}(\alpha)$*,*

$$|q_{\mathrm{mix}}(z) - q_\star(\alpha)| \le (1-\lambda)\epsilon/m + \lambda|q_{\mathrm{global}}(\alpha) - q_\star(\alpha)|. \tag{11}$$

The proof follows directly from the definition of the quantile and the assumed uniform CDF approximation. If the weighted CDF is within ϵ of the ideal local CDF, and the ideal CDF is not flat around its target quantile, the weighted quantile cannot move more than ϵ/m without crossing the target CDF level by more than the allowed CDF error. The second inequality follows by the triangle inequality. This result is best viewed as a diagnostic statement: it identifies the condition under which local weighting would be stable, but it does not prove that the proposed fingerprints satisfy that condition, nor does it replace the standard exchangeability condition required for split-conformal coverage.